

Is Multi-Person Gait Recognition Feasible under Mutual Occlusion? A Human Model Regression-based Approach

Ziruo Li Chi Xu* Xiang Li Shuqiong Wu Yasushi Yagi
 The University of Osaka, Osaka, Japan
 {z_li, xu, li, wu, yagi}@am.sanken.osaka-u.ac.jp

Abstract

Current gait recognition studies primarily focus on identifying a single person, assuming only one subject appears in the scene. However, in the wild, people often walk in groups, leading to mutual occlusion that degrades recognition performance. In this paper, we propose *Model-MultiGait*, a human model regression-based approach for multi-person gait recognition that explicitly accounts for mutual occlusion. Specifically, given a gait image sequence with multiple walking subjects, we directly regress parametric human models for all individuals, and extract gait features from each model to recognize them in an end-to-end manner. Unlike the standard pipeline that first detects each subject and then fits models independently, where mutual occlusion may cause missed detections and prevent model fitting for occluded individuals, our method simultaneously regresses multiple human models, capturing spatial and temporal relationships between subjects from a holistic perspective. Experiments on synthesized multi-person gait datasets demonstrate the effectiveness of the proposed method, achieving superior performance compared to state-of-the-art gait recognition methods.

1. Introduction

As a distinctive behavioral biometric in daily human walking, a subject's gait can be recognized without conscious awareness, making it difficult to imitate or conceal. Moreover, gait recognition remains effective even on low-resolution images captured from a distance, making it well-suited for applications such as criminal investigation, surveillance, and forensic analysis using CCTV footage [2, 25, 44]. Due to its strong application potential, gait recognition has received increasing attention over the past few decades [5, 12, 13, 19, 34, 40, 45, 51, 59, 60].

Most existing gait recognition studies primarily focus on recognizing a single person's gait, where only one walk-

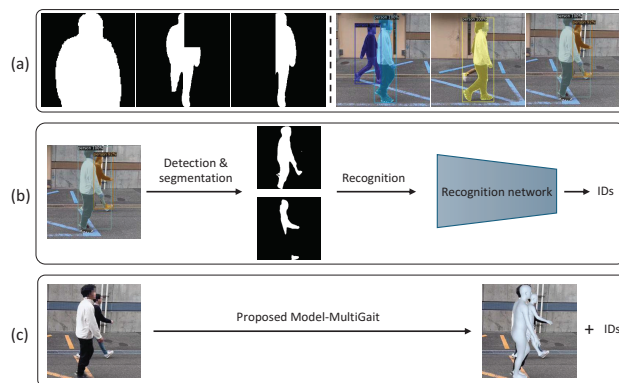


Figure 1. (a) Left: Simulated obstacle occlusion in existing gait recognition studies, which can be easily removed or replaced with a regular rectangle. Right: Real mutual occlusion in multi-person scenarios, often causing preprocessing errors, e.g., missed detection of the occluded subject (the one being blocked) in the second image, and segmentation errors merging the occluded subject's legs with the occluding subject (the one doing the blocking) in the third image. (b) Existing single-person pipeline-based multi-person gait recognition, which can be highly affected by occlusions and preprocessing errors. (c) The proposed Model-MultiGait, which simultaneously regresses human models for multiple subjects along with recognition results.

ing subject appears in the captured gait videos. However, in real-world applications, particularly in surveillance scenarios in the wild, people often walk in groups, such as with family, friends, or even strangers in crowded environments. When multiple individuals walk together, mutual occlusion between subjects is inevitable, affecting the completeness of gait feature representations and potentially leading to failures in preprocessing steps such as person detection, tracking, and segmentation, especially in cases of severe overlap. Although a few studies have explored multi-person gait recognition [7, 9], they follow a pipeline designed for single-person gait recognition, where each subject is detected and processed separately, without explicitly addressing the challenges posed by mutual occlusion (see Fig. 1(b)).

*Corresponding author.

On the other hand, some studies have addressed occlusion in single-person gait recognition [18, 47, 56, 62], where occlusion is typically simulated using rectangular obstacles, as shown in Fig. 1(a-left). In these cases, the obstruction can either be removed (e.g., the first image in Fig. 1(a-left)) or replaced with a black rectangle (e.g., the second and third images in Fig. 1(a-left)). However, in multi-person gait scenarios, mutual occlusion is far more complex. As illustrated in Fig. 1(a-right), it often involves irregular shapes, such as partial occlusion by another person’s torso or legs, and dynamically changes with movement and varying walking speeds, making it impossible to handle in the same way as rectangular occlusions.

Moreover, most existing occluded gait recognition methods rely on appearance-based representations (e.g., silhouettes), which are highly sensitive to errors in detection and instance segmentation. In multi-person settings, overlapping individuals can cause missed detections of occluded subjects (see the second image in Fig. 1(a-right)), leading to the complete omission of subsequent processing for those subjects. Segmentation failures may also occur, where parts of an occluded subject’s body remain in the occluding subject’s area (see the third image in Fig. 1(a-right)), introducing interference from another person’s gait. Consequently, these issues can significantly impact recognition performance, making single-person gait recognition pipelines unsuitable for handling multi-person gait videos effectively.

Compared to appearance-based representations, human model-based methods offer significant advantages in handling mutual occlusions in multi-person scenarios. Models like SMPL [42] impose structural and kinematic constraints, ensuring anatomically plausible body proportions and realistic joint movements. These constraints enable the estimation of a complete body even from limited visible parts through the model fitting process, while also helping to distinguish overlapping individuals based on body shape and posture, mitigating errors such as the inclusion of occluding body parts in an occluded subject’s gait representation. Notably, SMPL models have been successfully recovered for multiple persons in a single image, even under mutual occlusion [17, 27, 53]. At the same time, SMPL-based methods have demonstrated impressive performance in single-person gait recognition [35, 37, 62, 64, 71]. These advancements suggest that multi-person SMPL regression holds great potential for addressing the challenges of multi-person gait recognition, particularly in heavily occluded scenarios.

Motivated by these insights, we propose Model-MultiGait, a human model regression-based approach that simultaneously regresses multiple SMPL models to tackle mutual occlusion in multi-person gait recognition. The key contributions of this work are threefold.

1. End-to-end framework for multi-person gait recogni-

tion under mutual occlusion.

To our knowledge, this is the first end-to-end framework specifically designed for multi-person gait recognition that explicitly addresses mutual occlusion. Unlike existing approaches, which detect and recognize each subject independently, our method jointly processes multiple subjects in a unified pipeline, explicitly considering spatial and temporal interactions among individuals. The localization, gait feature extraction, and recognition stages are integrated and optimized in an end-to-end manner, substantially improving robustness under mutual occlusion.

2. Simultaneous multi-person gait modeling via human model regression.

The proposed Model-MultiGait concurrently regresses multiple SMPL models directly from input multi-person gait images (see Fig. 1(c)). Leveraging a Swin Transformer Encoder (STE) and a Multi-Person Regressor (MPR), our method reduces the impact of incomplete data and preprocessing errors encountered by single-person pipelines. Additionally, we introduce a location descriptor that enables the STE to explicitly encode spatial positions and interactions between subjects. Each regressed SMPL model is subsequently utilized for gait feature extraction, with joint training ensuring a balanced performance between human model regression and gait recognition.

3. State-of-the-art performance on multi-person gaits.

We evaluated the proposed method by synthesizing multi-person gait data using OU-MVLP [48]. To simulate real-world multi-person application scenarios, we generated various walking patterns with different degrees of mutual occlusion based on positional and viewpoint differences. Compared to state-of-the-art single-person pipeline-based gait recognition methods, our approach achieved significantly better recognition performance, particularly in cases of severe occlusion.

2. Related Work

2.1. Multi-Person Gait Recognition

Only a few studies have explored multi-person gait recognition [7, 9, 30, 52]. Chen et al. [7] used human graphlets for tracking and detection to segment silhouettes, then extracted gait dynamics from dense trajectories with attribute discovery via latent conditional random fields. Escano et al. [9] generated synthetic multi-person datasets from existing gait data, performing recognition by individually detecting subjects with Faster R-CNN [22], tracking them via ResNet50 features [20], and extracting gait features from optical flow maps through CNN-based methods [3, 4]. Recent works [30, 52] employed handcrafted features like HOG and linear kinematic attributes. However, all of these methods still approach multi-person gait recognition as independent single-person tasks, making them susceptible to

preprocessing errors from occlusion and overlapping subjects. In contrast, our proposed method explicitly performs joint multi-person modeling, effectively addressing these issues through an integrated end-to-end pipeline.

2.2. Single-Person Gait Recognition against Occlusion

While many single-person gait recognition studies address challenges in general application scenarios using either appearance-based representations, such as gait energy images (GEIs) [19, 33, 34, 51, 59, 60] and silhouettes [5, 12–14, 24, 40, 69], or model-based approaches leveraging skeletons [1, 15, 16, 39, 54, 55] and human meshes [35, 37, 64, 71], some studies specifically focus on occlusion, primarily using synthetic obstacle occlusion experiments. Most of these methods rely on GEIs or silhouettes, mitigating occlusion effects by extracting less-affected features [6, 26, 47, 58] or reconstructing complete gait images [21, 46, 50]. Compared to traditional machine learning approaches, CNN-based methods have demonstrated superior performance. Some methods use GANs to recover occluded silhouettes [8, 56], extract occlusion-invariant features via registration [63] or occlusion-awareness modules [18], and leverage action detection for dynamic contextual information [23]. A few model-based methods [61, 62] regress SMPL models from occluded gait videos. However, as previously mentioned, obstacle occlusion in single-person gait differs fundamentally from mutual occlusion in multi-person scenarios, making these methods unsuitable for direct application to multi-person gait recognition.

2.3. Multi-Person Human Mesh Recovery

Recent advances have significantly improved multi-person human mesh (e.g., SMPL) recovery. Zanfir et al. [66] used a feedforward framework with human and scene modeling to estimate SMPL parameters, later extending it to a bottom-up approach incorporating multi-person localization [67]. Jiang et al. [27] leveraged a Faster R-CNN [49]-based network with additional losses to penalize interpenetration and occlusions. Sun et al. [53] proposed a one-stage method that jointly predicts body center heatmaps and mesh parameters. More recently, Jiang et al. [28] introduced confidence-guided optimization with a layered neural representation and refined instance segmentation supervision. While most methods rely on CNN-based backbones, Goel et al. [17] proposed HMR 2.0, a fully transformer-based [11] approach, achieving superior performance over CNNs. Inspired by this, we adopt a transformer backbone for multi-SMPL regression in gait videos. However, unlike [17], which estimates SMPL independently for each person in a single image, our method regresses multiple SMPLs holistically, capturing both spatial and temporal relations across an image sequence.

3. Model-MultiGait

3.1. Overview

The overview framework of our method is shown in Fig. 2. As an initial step, we focus on gait images containing two walking persons. Given an input RGB sequence, we first pass it through the Swin Transformer Encoder (STE), which extracts latent features that capture both spatial and temporal relationships between subjects. These features, along with SMPL query tokens, are then fed into a Transformer decoder within the Multi-Person Regressor (MPR), which iteratively refines each subject’s camera, pose, and shape SMPL parameters via separate multilayer perceptron (MLP) heads. Simultaneously, a dedicated Location Estimator processes the latent features through additional MLP layers to generate a location descriptor for each subject. This descriptor explicitly encodes positional information, allowing STE to integrate locational and relational cues with image features, helping MPR distinguish overlapping individuals for more robust regression. Finally, the regressed SMPL parameters are used to extract individual gait features for identity recognition. The entire pipeline, encompassing both gait recognition and SMPL regression, is jointly optimized in an end-to-end manner through multiple loss functions, ensuring a balance between accurate recognition and consistent multi-person reconstruction.

3.2. Human Mesh Model

We adopt the SMPL model [42] to represent each subject’s 3D body configuration. SMPL decomposes the human body into pose parameters $\theta \in \mathbb{R}^{24 \times 6}$, which define the 6D rotation of 23 joints along with the global orientation, and shape parameters $\beta \in \mathbb{R}^{10}$, which capture body shape variations. Here, we use a 6D rotation representation instead of the default 3D axis-angle, as it is more suitable for regression according to [70]. Given θ and β , SMPL produces a triangulated mesh $\mathbf{M} \in \mathbb{R}^{6890 \times 3}$ and the corresponding 3D joint coordinates $\mathbf{J}_{3D}$. Additionally, we estimate a camera parameter $\mathbf{k} = [s, t_x, t_y] \in \mathbb{R}^3$, consisting of a 1D scale factor and 2D translation, to project 3D joints onto 2D joint locations in the image plane using a weak-perspective camera model. Compared to methods that rely purely on 2D silhouettes or heatmaps, this full 3D representation is more robust for gait recognition under occlusion, as each overlapping individual is recovered as a distinct mesh.

3.3. Swin Transformer Encoder

Instead of detecting and cropping each subject individually, the proposed method processes the entire multi-person input sequence holistically using an STE. Specifically, we adopt SwinV2-small [41] with a 16×16 window size, which is more parameter-efficient and better suited for sequential inputs compared to the ViT-H encoder used in HMR

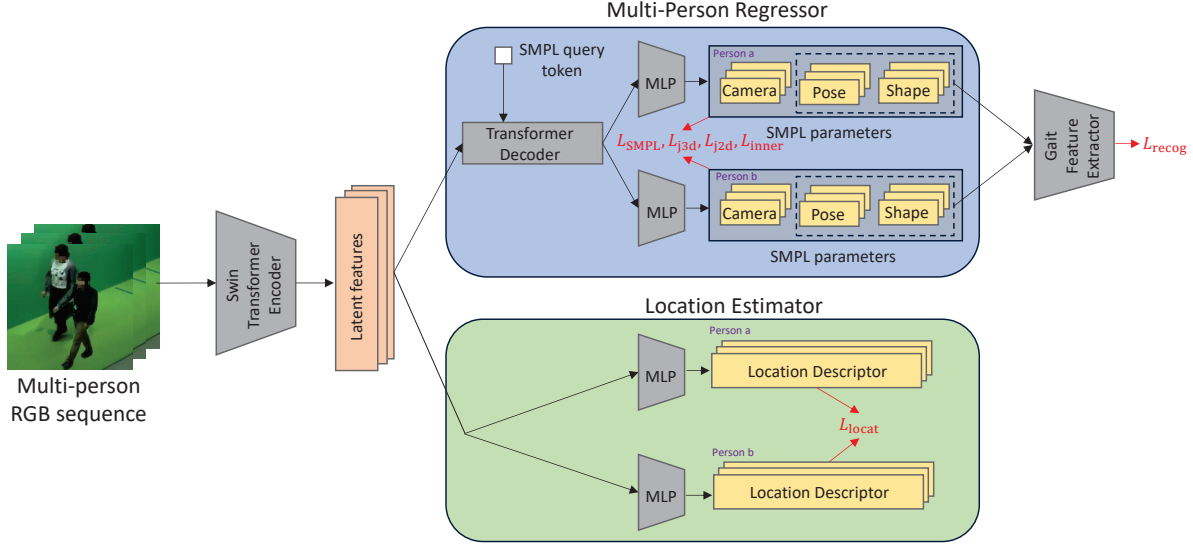


Figure 2. Framework of the proposed Model-MultiGait. Given an input multi-person RGB sequence containing persons a and b , latent features are first extracted by a Swin Transformer Encoder (STE). These features are then processed by two modules: (1) a Multi-Person Regressor (MPR), which regresses SMPL parameters for each individual, and (2) a Location Estimator, which generates location descriptors to encode positional information for each subject. The extracted SMPL shape and pose features are subsequently used for gait recognition.

2.0 [17]. The STE partitions each frame into patches, and applies hierarchical self-attention in shifted windows, effectively capturing both local details and global context. Since multiple people may appear in the same frame, the learned features encode relative spatial cues and visual representations of mutual occlusions, improving multi-person reconstruction accuracy.

Given an input sequence $S = \{s_1, s_2, \dots, s_n\}$, where n is the number of frames, the latent features are computed as

$$\mathbf{X}_i = \text{STE}(s_i), i \in \{1, 2, \dots, n\} \quad (1)$$

which retains hierarchical feature representations, preserving both local and global structural information for all visible subjects.

3.4. Multi-Person Regressor

After encoding, we employ a Transformer-based MPR to estimate SMPL parameters for each individual. The Transformer decoder follows the standard architecture with multi-head self-attention as in [17]. Given zero-valued SMPL query tokens $\mathbf{q}$ for each person, it applies cross-attention between $\mathbf{q}$ and the latent features $\mathbf{X}_i$, producing a structured representation that captures interactions among multiple individuals. Separate MLPs then regress the camera parameters $\mathbf{k}_i^p$, pose θ_i^p , and shape β_i^p for each person $p \in \{a, b\}$:

$$(\mathbf{k}_i^p, \theta_i^p, \beta_i^p) = \text{MLP}^p(\text{TransDec}(\mathbf{q}, \mathbf{X}_i)). \quad (2)$$

Since an individual's body shape remains consistent throughout the sequence, we apply average pooling to the

shape parameters across frames to obtain a mean shape representation for each subject.

As a result, the network predicts the SMPL parameters for all persons in a single forward pass. By iteratively refining these predictions, the model achieves better alignment with realistic human motion, effectively reducing ambiguities caused by partial overlaps, even in complex multi-person scenarios.

To train the MPR, we use a combination of SMPL losses, 3D joint losses, and 2D joint losses, following [17]. These losses ensure accurate pose and shape estimation while preserving spatial alignment with the image. Let the ground-truth (GT) SMPL parameters be $\Theta_{\text{gt}} = [\mathbf{k}_{\text{gt}}, \theta_{\text{gt}}, \beta_{\text{gt}}]$. We constrain the predictions using an MSE loss:

$$\mathcal{L}_{\text{SMPL}} = \|\Theta_{\text{pred}} - \Theta_{\text{gt}}\|_2^2, \quad (3)$$

where $\Theta_{\text{pred}} = [\mathbf{k}, \theta, \beta]$ represents the predicted SMPL parameters for each person.

Furthermore, let the GT 3D joint locations be $\mathbf{J}_{3D}^{\text{gt}}$ and GT 2D joint locations be $\mathbf{J}_{2D}^{\text{gt}}$. After obtaining the predicted 3D joints $\mathbf{J}_{3D}^{\text{pred}}$, we apply an L1 loss with respect to the GT 3D joints:

$$\mathcal{L}_{j3d} = \|\mathbf{J}_{3D}^{\text{pred}} - \mathbf{J}_{3D}^{\text{gt}}\|_1. \quad (4)$$

Similarly, after projecting the predicted joints onto the 2D image plane, we compute an L1 loss against the GT 2D joints:

$$\mathcal{L}_{j2d} = \|\mathbf{J}_{2D}^{\text{pred}} - \mathbf{J}_{2D}^{\text{gt}}\|_1. \quad (5)$$

Considering the temporal continuity of SMPL models in gait sequences, similar to [35], we introduce an addi-

tional inner loss, $\mathcal{L}_{\text{inner}}$, on the SMPL parameters. This loss enforces smooth transitions for camera and pose parameters between adjacent frames, as well as consistency for shape parameters across different sequences of the same subject, thereby integrating temporal coherence throughout the learning process.

3.5. Location Estimator

Typically, SMPL parameters are regressed using a single-person bounding box, where the camera parameter $\mathbf{k} = [s, t_x, t_y] \in \mathbb{R}^3$ is estimated relative to that box. However, this parameter alone does not represent the actual positions of multiple individuals within the full input image. To address this issue, we introduce a location descriptor $\mathbf{c} = [t_x^{\text{full}}, t_y^{\text{full}}, t_z^{\text{full}}] \in \mathbb{R}^3$, representing a 3D coordinate transformation with respect to the entire input image. Specifically, $\mathbf{c}_{\text{full}}$ is computed as in [32]:

$$\begin{aligned} t_x^{\text{full}} &= t_x + \frac{2(c_x - w/2)}{s * w_b}, & t_y^{\text{full}} &= t_y + \frac{2(c_y - h/2)}{s * h_b}, \\ t_z^{\text{full}} &= \frac{2f}{s * h_b}, \end{aligned} \quad (6)$$

where (c_x, c_y) , w_b , and h_b denote the bounding box center, width, and height, respectively; w and h represent the width and height of the input image; and f is the focal length. This descriptor implicitly captures bounding-box information, encoding each individual's spatial position and thus providing critical positional cues to the STE, which helps differentiate multiple subjects during regression.

To regress the location descriptor $\mathbf{c}_i^p$, we employ additional MLPs to process the latent features $\mathbf{X}_i$ learned by the STE, separately for each person $p \in \{a, b\}$:

$$\mathbf{c}_i^p = \text{MLP}^p(\mathbf{X}_i). \quad (7)$$

Finally, we supervise this prediction with an L1 loss:

$$\mathcal{L}_{\text{locat}} = \|\mathbf{c}^{\text{pred}} - \mathbf{c}^{\text{gt}}\|_1, \quad (8)$$

where $\mathbf{c}^{\text{pred}}$ and $\mathbf{c}^{\text{gt}}$ are the predicted and GT location descriptors, respectively.

3.6. Gait Feature Extractor

The regressed SMPL parameters are ultimately utilized for gait feature extraction. Following existing works [35, 62], we separately conduct gait recognition using two types of features: (1) shape-based recognition and (2) joint-based recognition.

Shape-based Recognition. Since the SMPL shape parameters explicitly encode body proportions such as height, weight, and overall build, they inherently serve as identity signatures for individuals. Therefore, we directly adopt the shape parameters $\mathbf{f}_{\text{shape}} = \beta$ as gait features for recognition.

Joint-based Recognition. Instead of directly employing the pose parameters (which represent joint rotations), we utilize the predicted 3D joint coordinates, following common practice in gait recognition. Specifically, we set the global orientation parameters to zero, producing human-centered 3D joint coordinates robust to viewpoint changes [36]. The per-frame 3D joint coordinates are flattened into a vector $\mathbf{j}_i \in \mathbb{R}^{72}$ and concatenated temporally across the sequence, forming a joint matrix representation $[\mathbf{j}_1, \mathbf{j}_2, \dots, \mathbf{j}_n] \in \mathbb{R}^{72 \times n}$. This representation is subsequently processed by a CNN with residual connections to generate a learned embedding $\mathbf{f}_{\text{joint}}$ for each sequence.

In both shape-based and joint-based pipelines, we adopt a triplet margin loss [57] for the recognition models:

$$\mathcal{L}_{\text{recog}} = \frac{1}{N} \sum_{i=1}^N \max(m + d_i^{\text{pos}} - d_i^{\text{neg}}, 0)^2, \quad (9)$$

where m is the margin, N is the number of triplets in a mini-batch, and d_i^{pos} and d_i^{neg} represent the Euclidean distances between embeddings of the positive pair and negative pairs for the i -th triplet, respectively. During training, this loss encourages the gait feature extractor to learn embeddings that remain consistent across sequences of the same subject while being discriminative across different individuals, even in scenarios involving mutual occlusion.

3.7. Joint Optimization

The entire framework is optimized jointly in an end-to-end manner by combining the previously defined loss functions, achieving an effective balance between SMPL regression accuracy and gait recognition performance. The overall training objective is formulated as follows:

$$\begin{aligned} \mathcal{L}_{\text{total}} &= w_{\text{SMPL}} \mathcal{L}_{\text{SMPL}} + w_{\text{j3d}} \mathcal{L}_{\text{j3d}} + w_{\text{j2d}} \mathcal{L}_{\text{j2d}} \\ &\quad + w_{\text{locat}} \mathcal{L}_{\text{locat}} + w_{\text{inner}} \mathcal{L}_{\text{inner}} + w_{\text{recog}} \mathcal{L}_{\text{recog}}, \end{aligned} \quad (10)$$

where w_{SMPL} , w_{j3d} , w_{j2d} , w_{locat} , w_{inner} , and w_{recog} represent weighting factors for their corresponding loss components.

4. Experiments

4.1. Dataset

Due to the lack of publicly available multi-person gait datasets, we constructed a synthetic dataset using the large-scale and multi-view OU-MVLP [48] as a foundation. Our synthetic dataset was generated through a combination of instance segmentation and image compositing techniques. Specifically, we first used PointRend [31] to segment the subjects from their original background. Each extracted subject (occluding subject) was then composited onto another subject's original image (occluded subject) captured

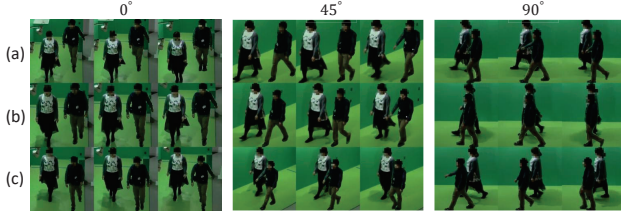


Figure 3. Examples images from our synthetic multi-person gait dataset: (a) “behind”; (b) “beside”; (c) “front”.

under the same view angle. To achieve realistic spatial overlap, we manually defined relative positional offsets for each viewpoint to maintain natural shoulder-to-shoulder spacing between subjects. By adjusting the offsets, we simulated three distinct walking patterns, i.e., “behind,” “beside,” and “front”, representing different positional relationships of the occluding subject relative to the occluded subject. The individual bounding boxes of both subjects were combined into a single bounding box, creating the input multi-person gait image.

To generate reliable pseudo-GT for training supervision, including location descriptors and SMPL parameters, we preserved corresponding single-person images aligned with the positions of each subject in the composited images, ensuring complete and unobstructed body information. Further details on the GT preparation are provided in the supplementary material.

We selected a subset of OU-MVLP containing 2,000 subjects (forming 1,000 subject pairs of multi-person gait images). Each pair includes three walking patterns and seven view angles (from 0° to 90° at 15° intervals), each with two sequences (“00”, “01”). The dataset was evenly split into training and testing subsets (500 subject pairs each). The “00” sequence serves as the probe set, while the “01” sequence acts as the gallery set. Example images from the synthetic multi-person gait dataset are shown in Fig. 3. Because the degree of occlusion varies depending on walking patterns and viewing angles, we calculated the mean occlusion ratio of the occluded subjects for each condition, as summarized in Fig. 4. The least occluded scenario occurs for the “behind” pattern at 0° (mean occlusion ratio 0.01%), whereas the most challenging scenarios occur for the “beside” pattern at 90° and the “front” pattern at 45° , with mean occlusion ratios of about 34.4% and 33.5%, respectively. Additionally, due to natural differences in walking speed, occlusion ratios vary across frames. For challenging cases, at least one-quarter of the frames within a sequence exhibit occlusion ratios exceeding 50%. Overall, this synthetic dataset provides a rigorous benchmark for evaluating multi-person gait recognition methods under realistic and diverse mutual occlusion conditions.

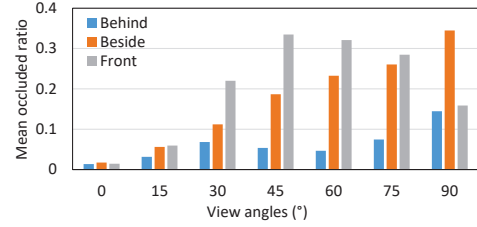


Figure 4. Mean occlusion ratio of the occluded subjects for each walking pattern and view angle.

4.2. Implementation Details

We resized the cropped multi-person gait images to 256×256 to match the input size of the SwinV2-small backbone [41], and uniformly sampled $n = 15$ frames per gait period. Training was conducted in two phases. In the first phase, we trained only the model regression modules (excluding the recognition loss) to achieve stable initial estimates. In the second phase, we incorporated the recognition module and the triplet margin loss, jointly optimizing the entire framework. The SwinV2-small weights were initialized from an ImageNet-1k pretrained model [10], while all other modules used default initialization. The detailed architecture of the proposed framework is provided in the supplementary material.

We trained our model using the AdamW optimizer [43] with a learning rate of $1e-5$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of $1e-4$. The batch size was set to 6×2 , corresponding to six subject pairs with two samples per pair in each mini-batch. The total number of training iterations was set to 60K, with the learning rate reduced to $1e-6$ after 30K iterations. Training hyperparameters remained identical for both phases. Inspired by loss weights in prior works [17, 35], we experimentally set the weights for each loss as follows: $w_{\text{SMPL}} = 1$, $w_{\text{j3d}} = 5$, $w_{\text{j2d}} = 1$, $w_{\text{locat}} = 0.1$, $w_{\text{inner}} = 1$, and $w_{\text{recog}} = 1$, respectively.

For evaluation, we adopted the widely used rank-1 identification rate and equal error rate (EER) to assess recognition performance under identification and verification scenarios, respectively.

4.3. Comparison with State-of-the-Arts

As mentioned, existing multi-person gait recognition methods [7, 9] essentially rely on single-person-based detection and segmentation pipeline. To provide a fair yet advanced baseline, we followed this pipeline using the widely adopted instance segmentation method PointRend [31] combined with multi-object tracking [68] to extract single-person gait sequences from our synthesized multi-person dataset as inputs for the comparison methods. Due to mutual occlusions, the extracted single-person sequences inevitably suffered from detection errors, resulting in an average loss of approximately five frames per sequence.

We compared our method against state-of-the-art

Table 1. Mean rank-1 rate [%] / EER [%] for the proposed method and comparison methods. Bold and bold italic indicate the best and second-best results, respectively.

Pipeline	Methods	Walking pattern			Subject role		Mean
		Behind	Beside	Front	Occluding	Occluded	
Single-person based	GaitBase [13]	73.52 / 2.83	61.87 / 5.24	59.96 / 5.68	72.37 / 2.69	55.67 / 6.74	64.34 / 4.77
	DeepGaitV2 [14]	73.65 / 2.75	62.06 / 4.82	60.05 / 5.27	72.17 / 2.75	56.08 / 6.13	64.43 / 4.46
	ModelGait [35]	58.99 / 2.81	50.72 / 4.07	48.10 / 4.30	57.06 / 2.67	45.25 / 5.02	51.36 / 3.83
Multi-person based	Ours-Pose	86.93 / 0.58	86.36 / 0.56	86.34 / 0.58	86.62 / 0.56	86.29 / 0.60	86.46 / 0.58
	Ours-Shape	90.36 / 0.27	90.28 / 0.27	90.11 / 0.31	90.25 / 0.28	90.25 / 0.29	90.25 / 0.29
	Ours-Fusion	90.35 / 0.27	90.29 / 0.26	90.12 / 0.29	90.26 / 0.28	90.25 / 0.27	90.26 / 0.27

(SOTA) appearance-based methods, GaitBase [13] and DeepGaitV2 [14], as well as the model-based method ModelGait [35]¹. GaitBase and DeepGaitV2 were implemented using the official OpenGait repository [13], and ModelGait followed the original paper’s default settings. For our method, we separately reported the results using shape features, pose features, and their fusion.

Given three walking patterns, each with seven viewing angles, the cross-pattern and cross-view evaluations resulted in 21×21 combinations. To simplify comparison, Table 1 summarizes (1) mean cross-view performance (7×7) within each pattern, (2) mean performance for occluding and occluded subjects across all 21×21 combinations, and (3) the overall average performance. Consistent with the occlusion difficulties shown in Fig. 4, the SOTA methods achieved significantly better performance on the minimally occluded “behind” pattern compared to the more challenging “beside” and “front” patterns. In contrast, our method’s results were more consistent across all three patterns, highlighting its robustness to varying degrees of occlusion. Moreover, our method exhibited the smallest performance gap between occluding and occluded subjects, indicating its effectiveness in capturing reliable gait features from severely occluded individuals. Consequently, our proposed end-to-end multi-person framework significantly outperformed all comparison methods, demonstrating its clear advantage in handling mutual occlusion.

Comparing variants within our method, we observed similar trends to those reported by ModelGait [35]: (1) shape-based features achieved better performance than pose-based features, and (2) applying score-level fusion further improved recognition accuracy.

4.4. Ablation Study

Ablation on loss functions. We evaluated the contributions of the SMPL inner loss $\mathcal{L}_{\text{inner}}$ and recognition loss $\mathcal{L}_{\text{recog}}$ by sequentially removing them, while retaining the reconstruction loss $\mathcal{L}_{\text{recons}}$, defined as the weighted combination of $\mathcal{L}_{\text{SMPL}}$, $\mathcal{L}_{\text{j3d}}$, $\mathcal{L}_{\text{j2d}}$, and $\mathcal{L}_{\text{locat}}$. Results are presented

¹We omitted OA-ModelGait [62], since its phase synchronization process requires sequences of at least half a gait cycle, which cannot always be guaranteed in heavily occluded scenarios.

Table 2. Overall mean rank-1 rate [%] / EER [%] of the proposed method using shape feature for the ablation experiments. “E2E” indicates the proposed end-to-end multi-person pipeline.

$\mathcal{L}_{\text{recons}}$	$\mathcal{L}_{\text{inner}}$	$\mathcal{L}_{\text{recog}}$	E2E	Backbone	Mean
✓	✓	×	×	STE+MPR	18.56 / 7.91
✓	×	✓	✓	STE+MPR	89.04 / 0.32
✓	✓	✓	×	STE+MPR	83.12 / 0.77
✓	✓	✓	✓	ResNet+MLP	75.79 / 1.09
✓	✓	✓	✓	STE+MPR	90.25 / 0.29

in the first two rows of Table 2. Removing $\mathcal{L}_{\text{recog}}$ significantly degraded performance, as relying solely on reconstruction loss provides insufficient discriminative features for recognition. Including $\mathcal{L}_{\text{inner}}$ slightly improved performance by enforcing temporal smoothness and shape consistency across sequences, thus benefiting gait recognition.

Ablation on end-to-end multi-person pipeline. We modified our method to regress only a single individual’s SMPL parameters, adapting it to the single-person gait data generated by the preprocessing pipeline described earlier. As shown in the third row of Table 2, the single-person pipeline led to notable performance degradation due to preprocessing errors caused by occlusions. This result highlights the effectiveness and necessity of our proposed end-to-end multi-person framework.

Ablation on backbone network. We replaced the proposed STE and MPR backbone with the ResNet and MLP-based regressor used in HMR [29]. This substitution led to a substantial drop in performance (fourth row of Table 2), clearly validating the effectiveness of our backbone choice.

4.5. Visualizing Multi-Person Models

We visualized the SMPL models regressed by our proposed method in Fig. 5. The results demonstrate that our method robustly estimates accurate SMPL models for both occluding and occluded individuals across various walking patterns and viewing angles. Even in challenging conditions such as “beside” at 90° , where only partial body regions of the occluded subject are visible, our method effectively reconstructs the full-body configuration by leveraging temporal context across frames. This capability directly enhances gait recognition performance under severe occlusion.

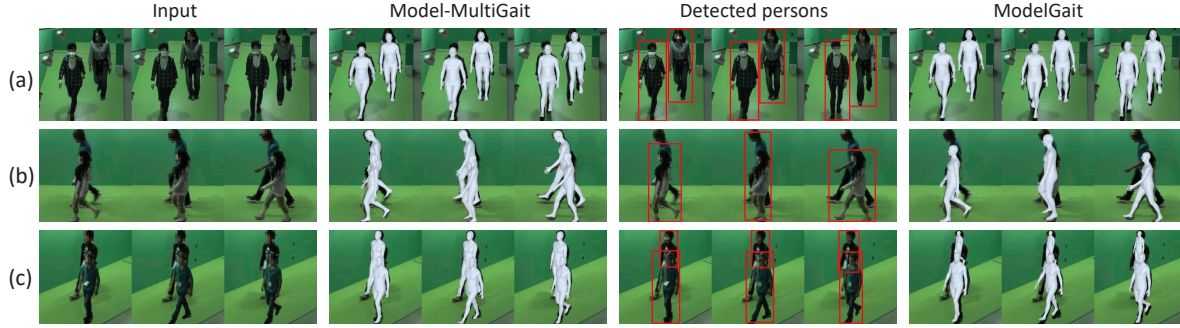


Figure 5. Examples of SMPL models estimated by the proposed Model-MultiGait and the single-person-based method ModelGait [35] under different scenarios: (a) “behind” (0°); (b) “beside” (90°); (c) “front” (45°). The third column shows the detected person bounding boxes using PointRender.

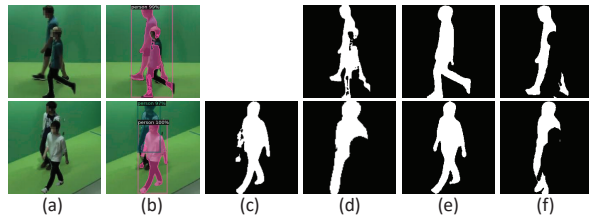


Figure 6. Comparison between segmentation results and GT silhouettes. Each row represents an example image. (a) Input image. (b) Instance segmentation results obtained via PointRender [31]. (c) and (e) show the segmented and GT silhouettes of the occluding subject, respectively. (d) and (f) show the segmented and GT silhouettes of the occluded subject, respectively.

Table 3. Overall mean rank-1 rate [%] / EER [%] for the proposed method and DeepGaitV2 [14], using silhouettes segmented by PointRender [31] or GT silhouettes.

Methods	DeepGaitV2-PointRender	DeepGaitV2-GT	Ours
Results	64.43 / 4.46	83.17 / 1.09	90.26 / 0.27

In comparison, the single-person pipeline (ModelGait [35]) suffers from noticeable detection errors, including missed or incorrect bounding boxes (third column of Fig. 5). Consequently, these errors lead to incomplete or incorrect SMPL estimations (fourth column), significantly reducing recognition reliability under severe occlusion scenarios.

5. Discussion

Effects of preprocessing errors in the single-person pipeline. To further analyze the lower performance of the single-person-based pipeline, we compared PointRender-segmented [31] silhouettes with GT silhouettes using two example frames with severe mutual occlusions. As shown in Fig. 6, failures in detecting the occluding subject (missing Fig. 6(c) in the first row) and segmentation errors in occluded subjects (see Fig. 6(d) in both rows) lead to sig-

nificant deviations from the GT. Applying DeepGaitV2 [14] to GT silhouettes for all subjects substantially improves recognition performance (see Table 3), establishing an upper bound for the current single-person pipeline. However, due to occlusions, it still underperforms the proposed multi-person framework by approximately 7% in rank-1 rate and 0.8% in EER, highlighting the advantages of end-to-end multi-subject processing.

Limitations. While this study focuses on simulated scenarios with two walking individuals, future work will involve collecting real-world data with varying numbers of subjects. In such cases, the current framework, which regresses a fixed number of SMPL models, may be inadequate. A possible extension is to incorporate a flexible strategy for estimating both the probability and location of human presence in an image, enabling adaptive processing of varying numbers of individuals in real-world scenarios.

6. Conclusion

In this paper, we propose an end-to-end model regression-based approach for multi-person gait recognition, addressing the challenges posed by mutual occlusion. Given a video sequence containing multiple individuals, our method simultaneously regresses SMPL models for all subjects through a combination of STE, MPR, and Location Estimator. The regressed models are then utilized for gait feature extraction to identify each subject in the sequence. Experiments on synthesized multi-person gait data demonstrate that our approach achieves state-of-the-art performance. For future work, we plan to collect real-world multi-person walking data to capture more complex and realistic scenarios. Additionally, we aim to extend the framework to dynamically handle varying numbers of subjects, facilitating its application to diverse and flexible in-the-wild settings.

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